



PHYSICS
STANDARD LEVEL
PAPER 2

Topic A - Mechanics Study Guide

May 2009 - 2013

1 hour 15 minutes

Candidate session number

0	0							
---	---	--	--	--	--	--	--	--

INSTRUCTIONS TO CANDIDATES

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Section A: answer all of Section A in the spaces provided.
- Section B: answer one question from Section B in the spaces provided.
- At the end of the examination, indicate the numbers of the questions answered in the candidate box on your cover sheet.



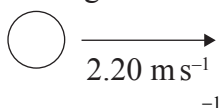
A2. This question is about impulse.

- (a) A net force of magnitude F acts on a body. Define the *impulse* I of the force. [1]

.....

- (b) A ball of mass 0.0750 kg is travelling horizontally with a speed of 2.20 m s^{-1} . It strikes a vertical wall and rebounds horizontally.

ball mass
 0.0750 kg



Due to the collision with the wall, 20% of the ball's initial kinetic energy is dissipated.

- (i) Show that the ball rebounds from the wall with a speed of 1.97 m s^{-1} . [2]

.....

- (ii) Show that the impulse given to the ball by the wall is 0.313 N s . [2]

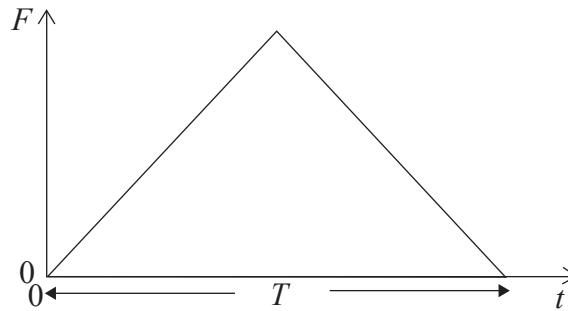
.....

(This question continues on the following page)

(Question A2 continued)

- (c) The ball strikes the wall at time $t = 0$ and leaves the wall at time $t = T$.

The sketch graph shows how the force F that the wall exerts on the ball is assumed to vary with time t .



The time T is measured electronically to equal 0.0894 s.

Use the impulse given in (b)(ii) to estimate the average value of F .

[4]

.....

.....

.....

.....

.....

.....

(Question B1 continued)

Part 2 Motion of a ball falling in oil

- (a) Distinguish between average speed and instantaneous speed. [2]

.....

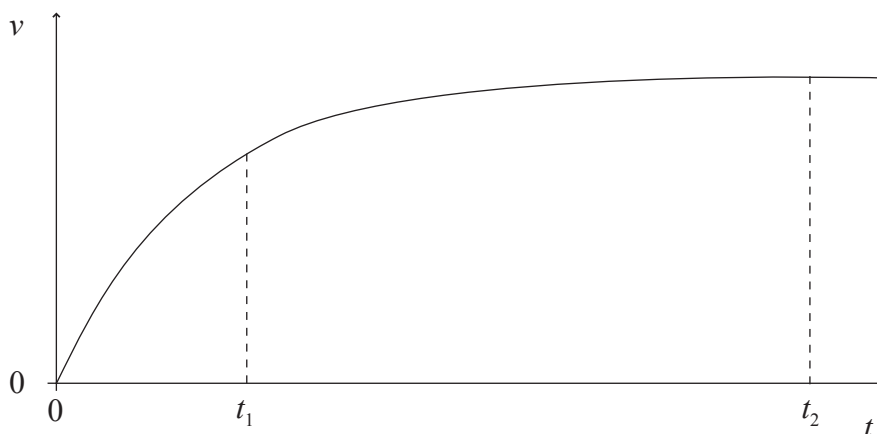
.....

.....

.....

- (b) A small steel ball of mass M is dropped from rest into a long vertical tube that contains oil.

The sketch graph shows how the speed v of the ball varies with time t .



- Explain how you would use the graph to find the average speed of the ball between $t = 0$ and $t = t_1$. [3]

.....

.....

.....

.....

.....

(This question continues on the following page)

(Question B1, part 2 continued)

- (c) The gradient of the graph at $t = t_1$ is k . Deduce an expression in terms of k , M and g , the acceleration of free fall, for the magnitude of the frictional force F acting on the ball at $t = t_1$. [3]

.....

.....

.....

.....

.....

- (d) State and explain the magnitude of the frictional force acting on the ball at $t = t_2$. [3]

.....

.....

.....

.....

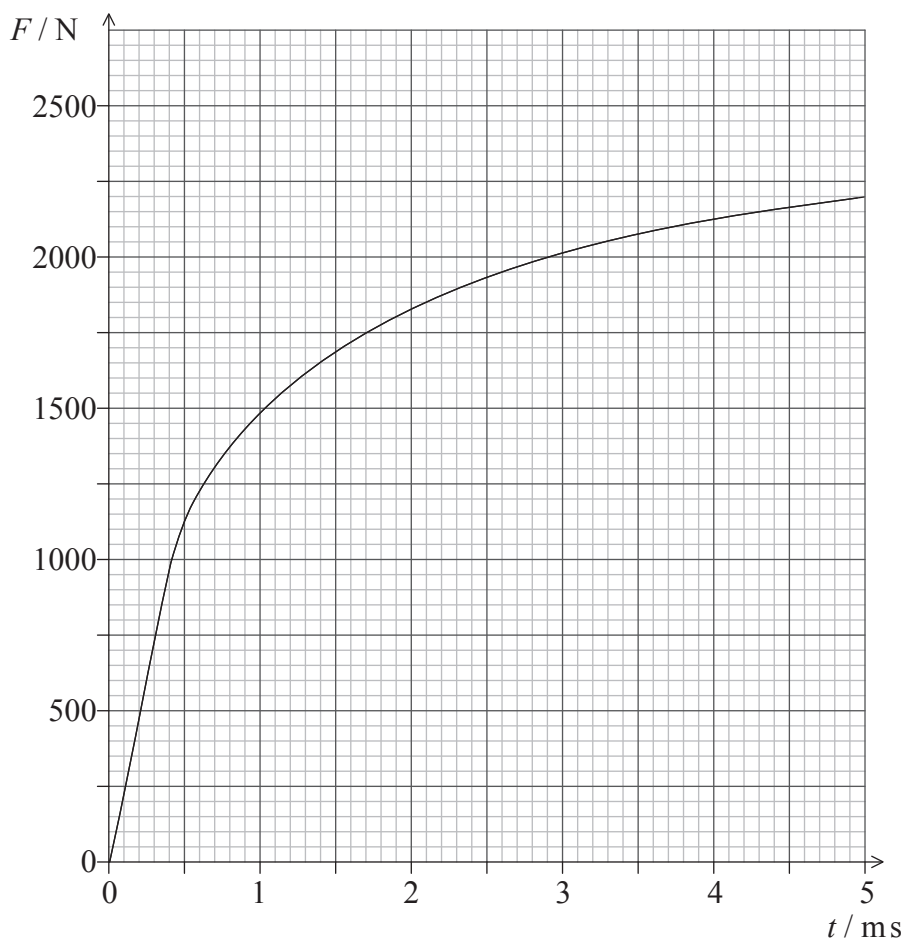
SECTION B

This section consists of three questions: B1, B2 and B3. Answer **one** question.

B1. This question is in **two** parts. **Part 1** is about dynamics and energy. **Part 2** is about the use of fossil fuels.

Part 1 Dynamics and energy

A bullet of mass 32 g is fired from a gun. The graph shows the variation of the force F on the bullet with time t as it travels along the barrel of the gun.



The bullet is fired at time $t=0$ and the length of the barrel is 0.70 m.

(This question continues on the following page)

(Question B1, part 1 continued)

- (a) State and explain why it is inappropriate to use the equation $s = ut + \frac{1}{2}at^2$ to calculate the acceleration of the bullet. [2]

.....

.....

.....

.....

.....

- (b) Use the graph to

- (i) determine the average acceleration of the bullet during the final 2.0 ms of the graph. [2]

.....

.....

.....

.....

.....

- (ii) show that the change in momentum of the bullet, as the bullet travels along the length of the barrel, is approximately 9 N s. [3]

.....

.....

.....

.....

.....

(This question continues on the following page)

(Question B1, part 1 continued)

(c) Use the answer in (b)(ii) to calculate the

(i) speed of the bullet as it leaves the barrel. [2]

.....
.....
.....
.....
.....

(ii) average power delivered to the bullet. [3]

.....
.....
.....
.....
.....

(d) Use Newton’s third law to explain why a gun will recoil when a bullet is fired. [3]

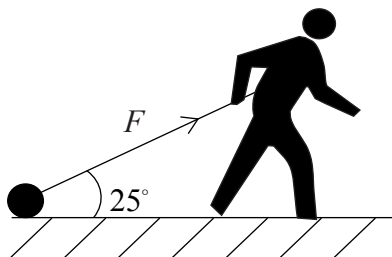
.....
.....
.....
.....
.....

(This question continues on page 14)



A2. This question is about forces.

An athlete trains by dragging a heavy load across a rough horizontal surface.



The athlete exerts a force of magnitude F on the load at an angle of 25° to the horizontal.

- (a) Once the load is moving at a steady speed, the average horizontal frictional force acting on the load is 470 N.

Calculate the average value of F that will enable the load to move at constant speed. [2]

.....

.....

.....

.....

- (b) The load is moved a horizontal distance of 2.5 km in 1.2 hours.

Calculate

- (i) the work done on the load by the force F . [2]

.....

.....

.....

- (ii) the minimum average power required to move the load. [2]

.....

.....

.....

- (c) The athlete pulls the load uphill at the same speed as in part (a).

Explain, in terms of energy changes, why the minimum average power required is greater than in (b)(ii). [2]

.....

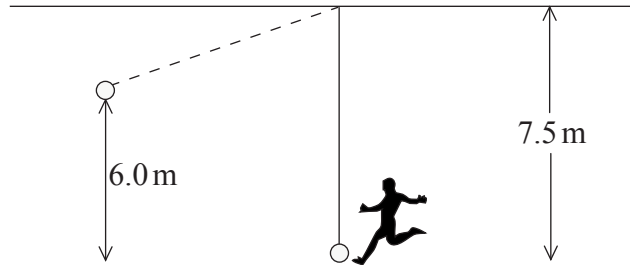
.....

.....

(Question B1 continued)

Part 2 Kicking a football

A ball is suspended from a ceiling by a string of length 7.5 m. The ball is kicked horizontally and rises to a maximum height of 6.0 m.



- (a) Assuming that the air resistance is negligible, show that the initial speed of the ball is 11 ms^{-1} . [2]

.....

- (b) The mass of the ball is 0.55 kg and the impact time of the kicker's foot with the ball is 150 ms. Estimate the average force exerted on the ball by the kick. [2]

.....

- (c) (i) Explain why the tension in the string increases immediately after the ball is kicked. [3]

.....

- (ii) Calculate the tension in the string immediately after the ball is kicked. Assume that the string is vertical. [3]

.....

B3. This question is in **two** parts. **Part 1** is about momentum, energy and power. **Part 2** is about electric and gravitational fields.

Part 1 Momentum, energy and power

- (a) In his *Principia Mathematica* Newton expressed his third law of motion as “to every action there is always opposed an equal reaction”. State what Newton meant by this law. [1]

.....

.....

.....

- (b) A book is released from rest and falls towards the surface of Earth. Discuss how the conservation of momentum applies to the Earth-book system. [3]

.....

.....

.....

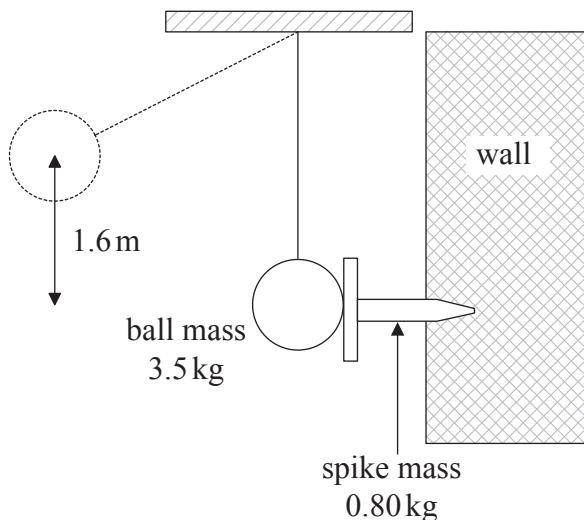
.....

.....

(This question continues on the following page)

(Question B3, part 1 continued)

- (c) A large swinging ball is used to drive a horizontal iron spike into a vertical wall. The centre of the ball falls through a vertical height of 1.6 m before striking the spike in the position shown.



The mass of the ball is 3.5 kg and the mass of the spike is 0.80 kg. Immediately after striking the spike, the ball and spike move together. Show that the

- (i) speed of the ball on striking the spike is 5.6 m s^{-1} . [1]

.....

- (ii) energy dissipated as a result of the collision is about 10 J. [4]

.....

(This question continues on the following page)

(Question B3, part 1 continued)

- (d) As a result of the ball striking the spike, the spike is driven a distance $7.3 \times 10^{-2} \text{ m}$ into the wall. Calculate, assuming it to be constant, the friction force F between the spike and wall. [3]

.....

.....

.....

.....

.....

- (e) The machine that is used to raise the ball has a useful power output of 18 W. Calculate how long it takes for the machine to raise the ball through a height of 1.6 m. [3]

.....

.....

.....

.....

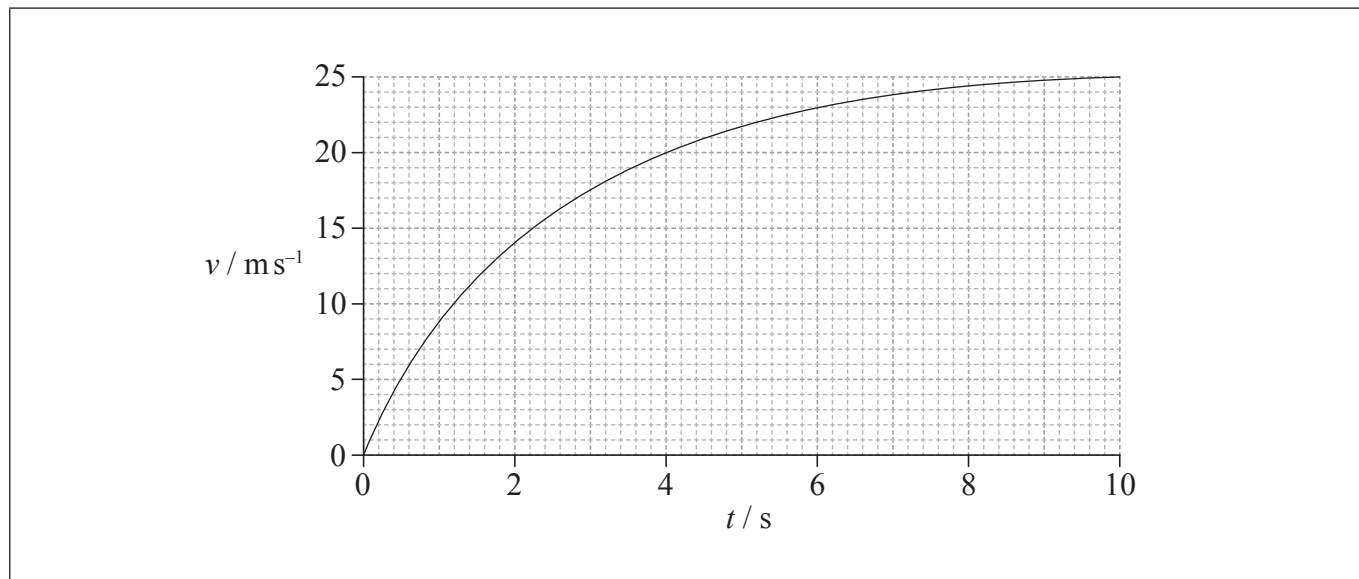
.....

(This question continues on the following page)

B2. This question is in **two** parts. **Part 1** is about mechanics and thermal physics. **Part 2** is about nuclear physics.

Part 1 Mechanics and thermal physics

The graph shows the variation with time t of the speed v of a ball of mass 0.50 kg , that has been released from rest above the Earth's surface.



The force of air resistance is **not** negligible. Assume that the acceleration of free fall is $g = 9.81\text{ ms}^{-2}$.

- (a) State, without any calculations, how the graph could be used to determine the distance fallen. [1]

.....

.....

(This question continues on the following page)



(Question B2, part 1 continued)

- (b) (i) In the space below, draw and label arrows to represent the forces on the ball at 2.0 s. [1]

ball at
 $t = 2.0 \text{ s}$ ●

Earth's surface _____

- (ii) Use the graph opposite to show that the acceleration of the ball at 2.0 s is approximately 4 ms^{-2} . [2]

.....

.....

.....

.....

- (iii) Calculate the magnitude of the force of air resistance on the ball at 2.0 s. [2]

.....

.....

.....

.....

(This question continues on the following page)



(Question B2, part 1 continued)

- (iv) State and explain whether the air resistance on the ball at $t=5.0\text{ s}$ is smaller than, equal to **or** greater than the air resistance at $t=2.0\text{ s}$. [2]

.....

(This question continues on the following page)



(Question B2, part 1 continued)

(c) After 10 s the ball has fallen 190 m.

- (i) Show that the sum of the potential and kinetic energies of the ball has decreased by 780 J. [3]

.....

.....

.....

.....

.....

.....

- (ii) The specific heat capacity of the ball is $480 \text{ J kg}^{-1} \text{ K}^{-1}$. Estimate the increase in the temperature of the ball. [2]

.....

.....

.....

.....

- (iii) State an assumption made in the estimate in (c)(ii). [1]

.....

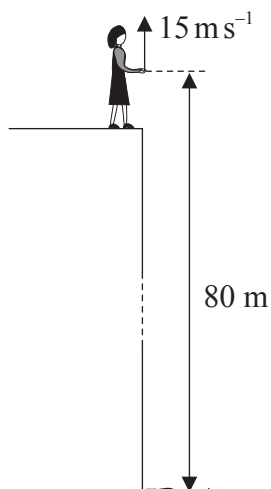
.....

(This question continues on the following page)



A2. This question is about kinematics.

Lucy stands on the edge of a vertical cliff and throws a stone vertically upwards.



The stone leaves her hand with a speed of 15 m s^{-1} at the instant her hand is 80 m above the surface of the sea. Air resistance is negligible and the acceleration of free fall is 10 m s^{-2} .

- (a) Calculate the maximum height reached by the stone as measured from the point where it is thrown. [2]

.....

.....

.....

.....

- (b) Determine the time for the stone to reach the surface of the sea after leaving Lucy's hand. [3]

.....

.....

.....

.....

.....

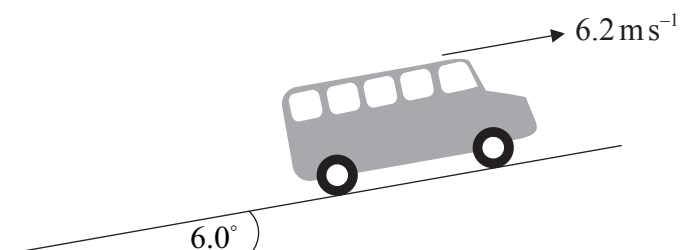
.....



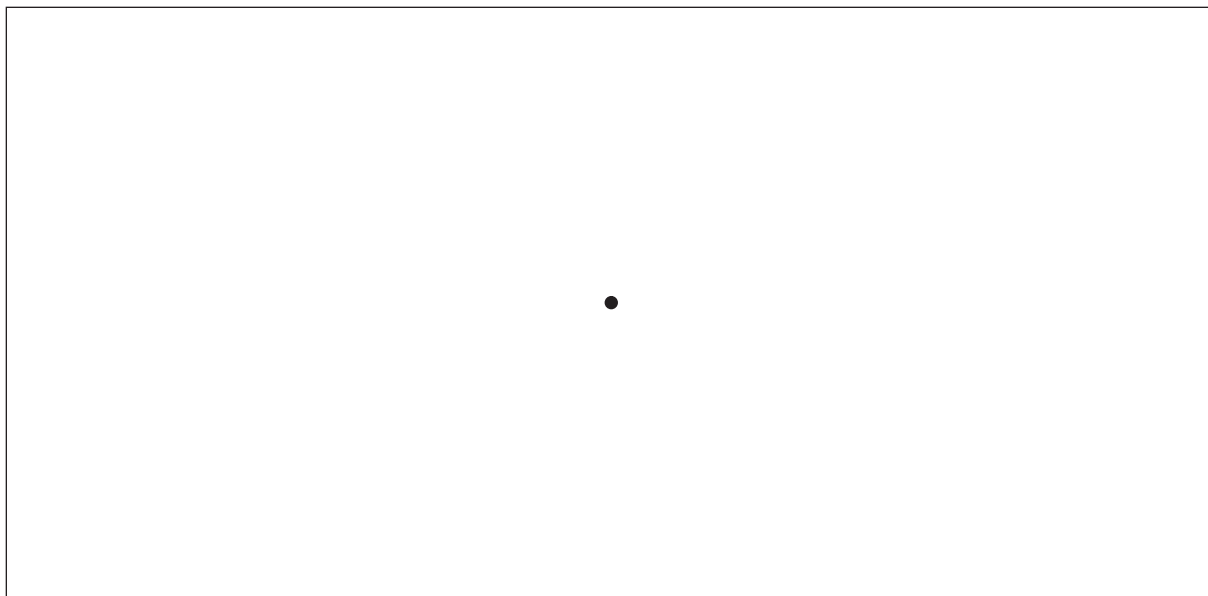
- B3.** This question is in **two** parts. **Part 1** is about power and efficiency. **Part 2** is about electrical resistance.

Part 1 Power and efficiency

A bus is travelling at a constant speed of 6.2 ms^{-1} along a section of road that is inclined at an angle of 6.0° to the horizontal.



- (a) (i) The bus is represented by the black dot shown below. Draw a labelled sketch to represent the forces acting on the bus. [4]



- (ii) State the value of the rate of change of momentum of the bus. [1]

.....

(This question continues on the following page)



(Question B3, part 1 continued)

- (b) The total output power of the engine of the bus is 70 kW and the efficiency of the engine is 35 %. Calculate the input power to the engine. [2]

.....

.....

.....

.....

- (c) The mass of the bus is 8.5×10^3 kg. Determine the rate of increase of gravitational potential energy of the bus. [3]

.....

.....

.....

.....

.....

- (d) Using your answer to (c) and the data in (b), estimate the magnitude of the resistive forces acting on the bus. [3]

.....

.....

.....

.....

.....

(This question continues on the following page)



(Question B3, part 1 continued)

(e) The engine of the bus suddenly stops working.

- (i) Determine the magnitude of the net force opposing the motion of the bus at the instant at which the engine stops. [2]

.....

.....

.....

.....

- (ii) Discuss, with reference to the air resistance, the change in the net force as the bus slows down. [2]

.....

.....

.....

.....

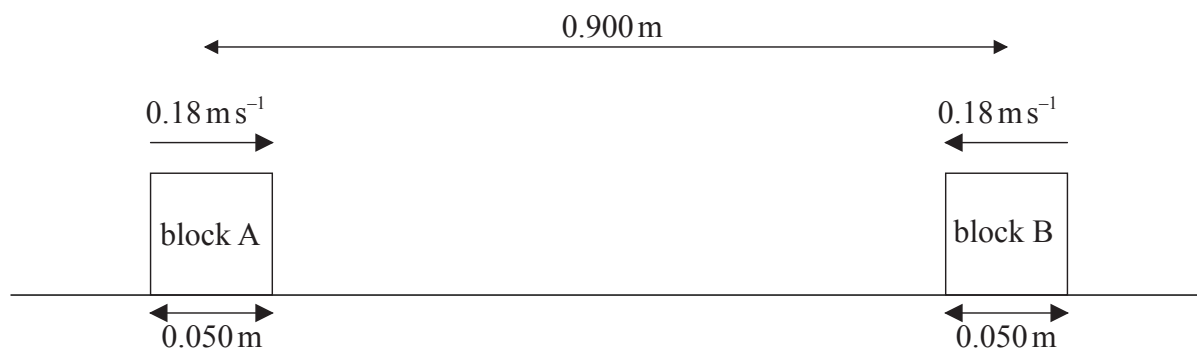
(This question continues on the following page)



- B3.** This question is in **two** parts. **Part 1** is about a collision. **Part 2** is about electric current and resistance.

Part 1 A collision

Two identical blocks of mass 0.17 kg and length 0.050 m are travelling towards each other along a straight line through their centres as shown below. Assume that the surface is frictionless.



The initial distance between the centres of the blocks is 0.900 m and both blocks are moving at a speed of 0.18 ms^{-1} relative to the surface.

- (a) Determine the time taken for the blocks to come into contact with each other. [3]

.....

.....

.....

.....

.....

.....

(This question continues on the following page)



(Question B3, part 1 continued)

- (b) As a result of the collision, the blocks reverse their direction of motion and travel at the same speed as each other. During the collision, 20 % of the kinetic energy of the blocks is given off as thermal energy to the surroundings.

- (i) State and explain whether the collision is elastic or inelastic. [2]

.....

.....

.....

.....

- (ii) Show that the final speed of the blocks relative to the surface is 0.16 m s^{-1} . [3]

.....

.....

.....

.....

.....

.....

(This question continues on the following page)



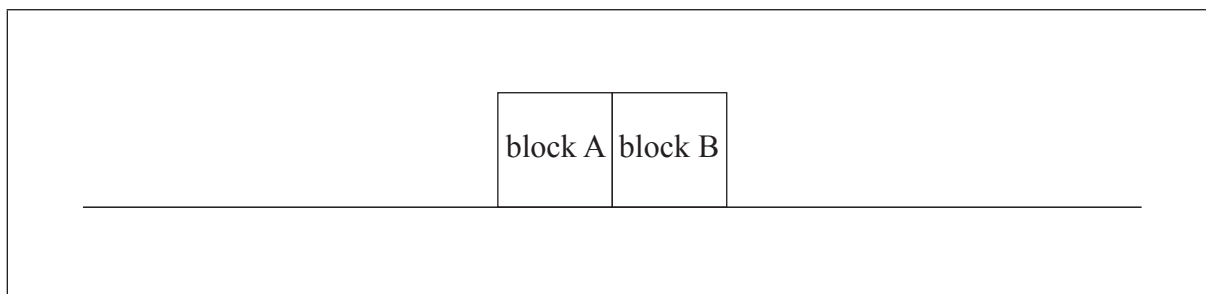
(Question B3, part 1 continued)

- (c) (i) State Newton's third law of motion. [1]

.....

.....

- (ii) During the collision of the blocks, the magnitude of the force that block A exerts on block B is F_{AB} and the magnitude of the force that block B exerts on block A is F_{BA} . On the diagram below, draw labelled arrows to represent the magnitude and direction of the forces F_{AB} and F_{BA} . [3]



- (iii) The duration of the collision between the blocks is 0.070 s. Determine the average force one block exerted on the other. [3]

.....

.....

.....

.....

.....

.....

(This question continues on page 26)



A2. This question is about kinematics.

(a) State the difference between average speed and instantaneous speed.

[2]

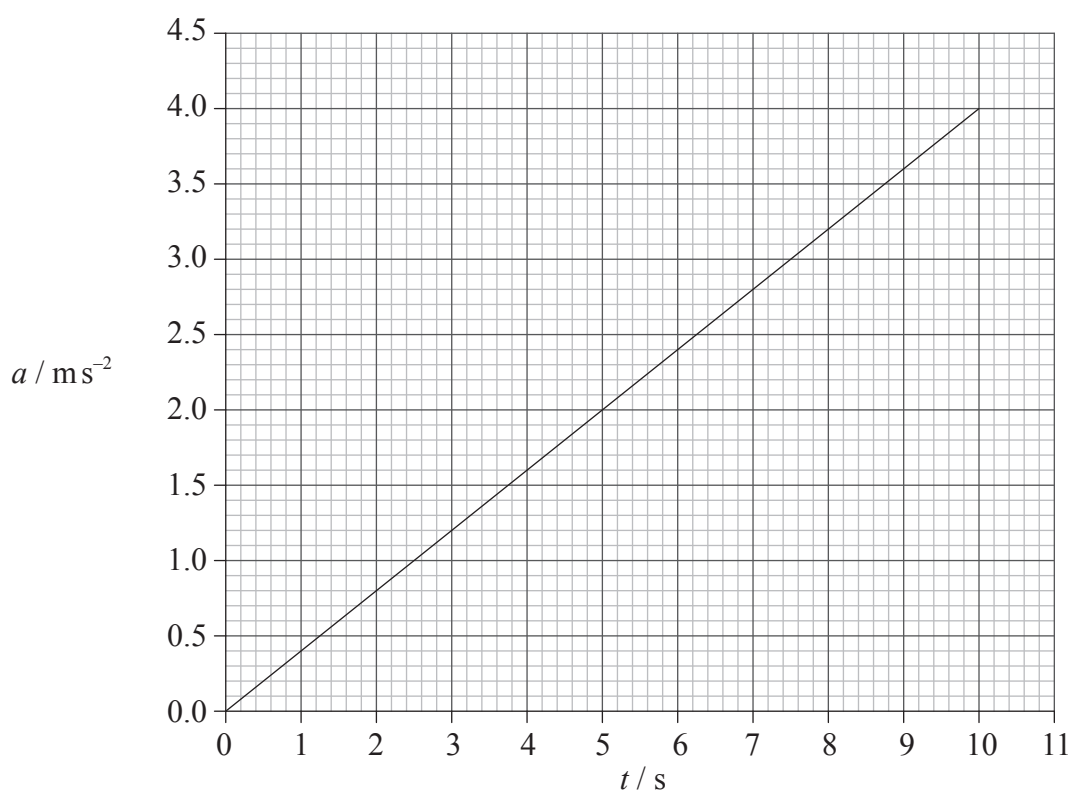
.....

.....

.....

.....

(b) The graph shows how the acceleration a of a particle varies with time t .



At time $t=0$ the instantaneous speed of the particle is zero.

(This question continues on the following page)



(Question A2 continued)

- (i) Calculate the instantaneous speed of the particle at $t = 7.5$ s. [2]

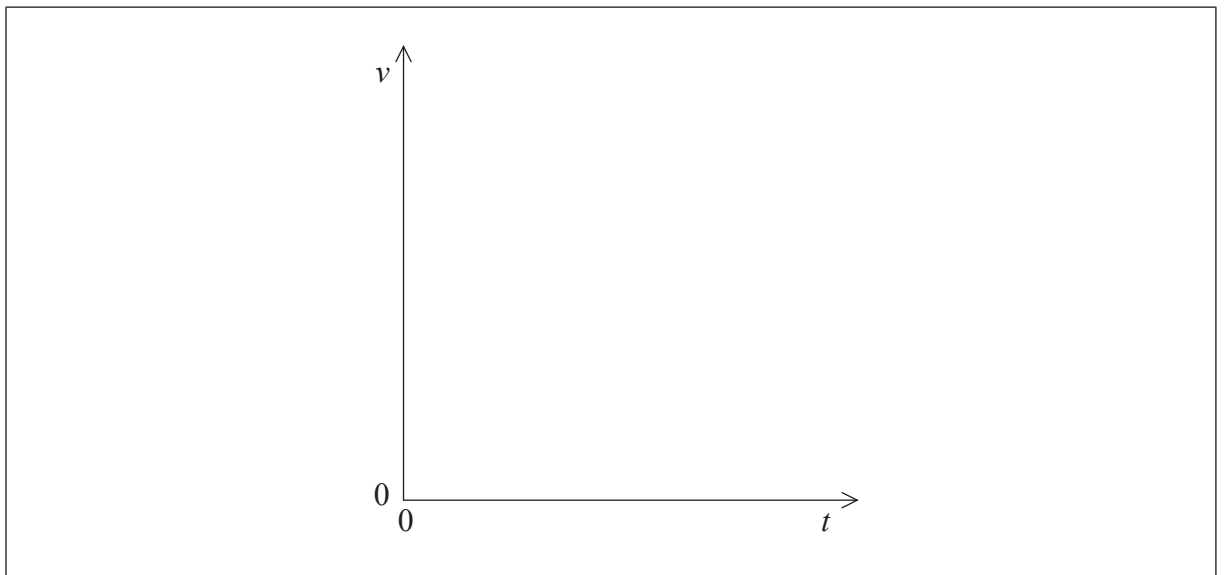
.....

.....

.....

.....

- (ii) Using the axes below, sketch a graph to show how the instantaneous speed v of the particle varies with t . [1]



- B2.** This question is in **two** parts. **Part 1** is about kinematics and mechanics. **Part 2** is about electric potential difference and electric circuits.

Part 1 Kinematics and mechanics

- (a) Define *linear momentum*. [1]

.....

.....

- (b) State, in terms of momentum, Newton's second law of motion. [1]

.....

.....

- (c) Show, using your answer to (b), how the impulse of a force F is related to the change in momentum Δp that it produces. [1]

.....

.....

.....

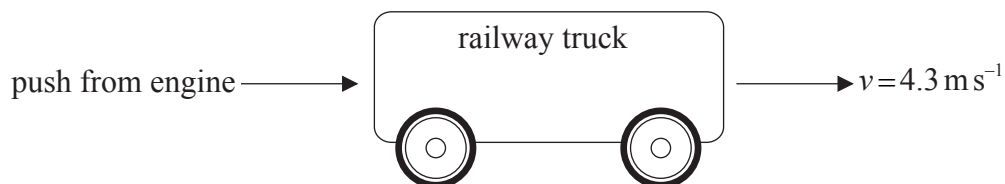
.....

(This question continues on the following page)



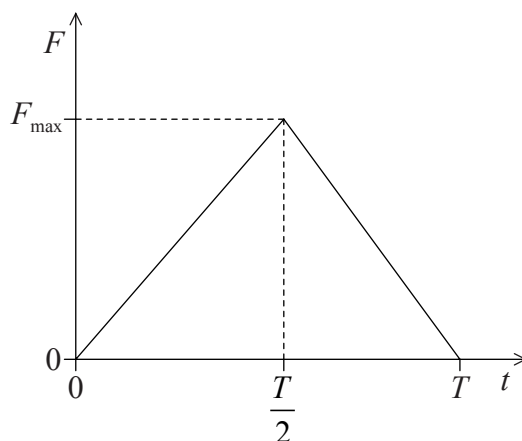
(Question B2, part 1 continued)

- (d) A railway truck on a level, straight track is initially at rest. The truck is given a quick, horizontal push by an engine so that it now rolls along the track.



The engine is in contact with the truck for a time $T = 0.54 \text{ s}$ and the initial speed of the truck after the push is 4.3 m s^{-1} . The mass of the truck is $2.2 \times 10^3 \text{ kg}$.

Due to the push, a force of magnitude F is exerted by the engine on the truck. The sketch shows how F varies with contact time t .



- (i) Determine the magnitude of the maximum force F_{max} exerted by the engine on the truck. [4]

.....

.....

.....

.....

.....

.....

.....

.....

(This question continues on the following page)



(Question B2, part 1 continued)

- (ii) After contact with the engine ($t=0.54\text{ s}$) the truck moves a distance 15 m along the track. After travelling this distance the speed of the truck is 2.8 m s^{-1} . Assuming a uniform acceleration, calculate the time it takes the truck to travel 15 m . [2]

.....

.....

.....

.....

- (iii) Calculate the average rate at which the kinetic energy of the truck is dissipated as it moves along the track. [2]

.....

.....

.....

.....

- (iv) When the speed of the truck is 2.8 m s^{-1} it collides with a stationary truck of mass $3.0 \times 10^3\text{ kg}$. The two trucks move off together with a speed V . Show that the speed $V=1.2\text{ m s}^{-1}$. [2]

.....

.....

.....

.....

(This question continues on the following page)



(Question B2, part 1 continued)

- (v) Outline the energy transformations that take place during the collision of the two trucks.

[2]

.....
.....
.....
.....

(This question continues on the following page)



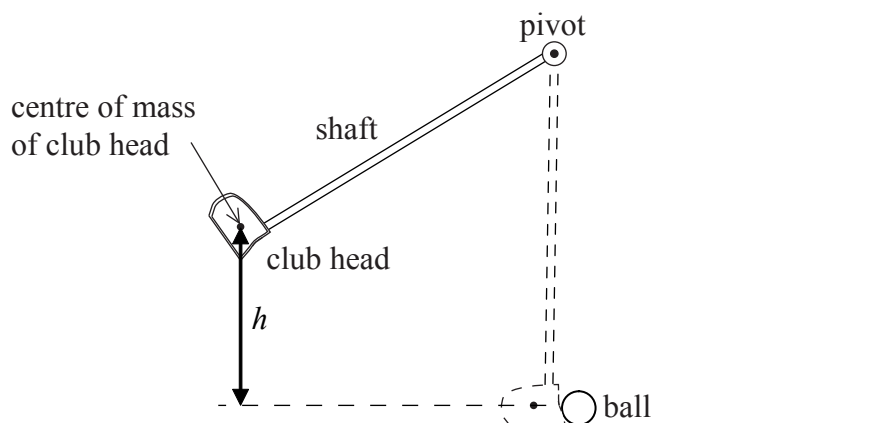
SECTION B

This section consists of three questions: B1, B2 and B3. Answer **one** question. Write your answers in the boxes provided.

- B1.** This question is in **two** parts. **Part 1** is about impulse and momentum. **Part 2** is about nuclear fusion.

Part 1 Impulse and momentum

The diagram shows an arrangement used to test golf club heads.



The shaft of a club is pivoted and the centre of mass of the club head is raised by a height h before being released. On reaching the vertical position the club head strikes the ball.

- (a) (i) Describe the energy changes that take place in the club head from the instant the club is released until the club head and the ball separate. [2]

.....

.....

.....

.....

(This question continues on the following page)



(Question B1, part 1 continued)

- (ii) Calculate the maximum speed of the club head achievable when $h = 0.85$ m. [2]

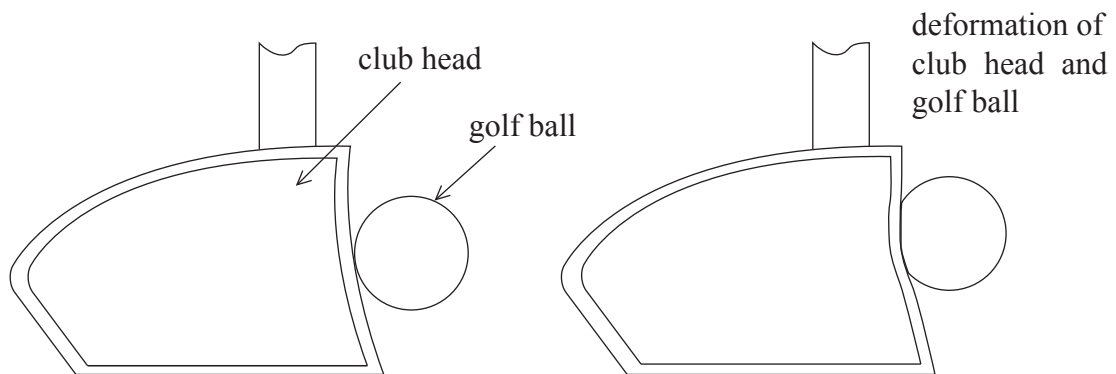
.....

.....

.....

.....

- (b) The diagram shows the deformation of a golf ball and club head as they collide during a test.



Explain how increasing the deformation of the club head may be expected to increase the speed at which the ball leaves the club. [2]

.....

.....

.....

.....

(This question continues on the following page)



(Question B1, part 1 continued)

- (c) In a different experimental arrangement, the club head is in contact with the ball for a time of $220\ \mu\text{s}$. The club head has mass $0.17\ \text{kg}$ and the ball has mass $0.045\ \text{kg}$. At the moment of contact the ball is at rest and the club head is moving with a speed of $38\ \text{m s}^{-1}$. The ball moves off with an initial speed of $63\ \text{m s}^{-1}$.

- (i) Calculate the average force acting on the ball while the club head is in contact with the ball. [2]

.....

.....

.....

.....

- (ii) State the average force acting on the club head while it is in contact with the ball. [1]

.....

.....

- (iii) Calculate the speed of the club head at the instant that it loses contact with the ball. [2]

.....

.....

.....

.....

(This question continues on page 14)



A2. This question is about kinematics.

- (a) Fiona drops a stone from rest vertically down a water well. She hears the splash of the stone striking the water 1.6 s after the stone leaves her hand. Estimate the

- (i) distance between Fiona’s hand and the water surface.

[1]

.....
.....

- (ii) speed with which the stone hits the water.

[2]

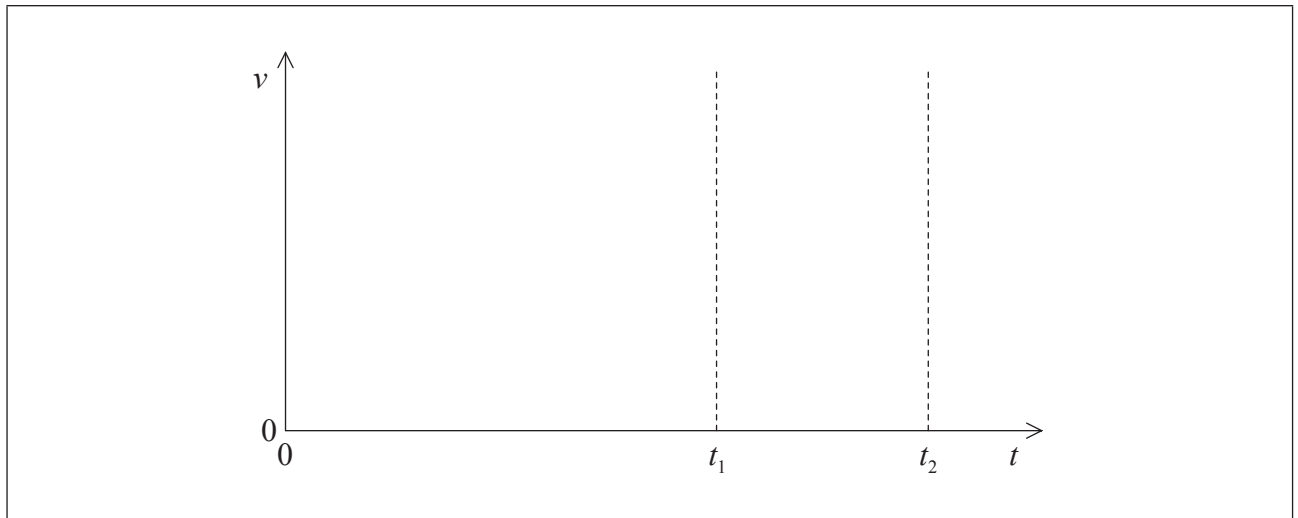
.....
.....
.....
.....

(This question continues on the following page)

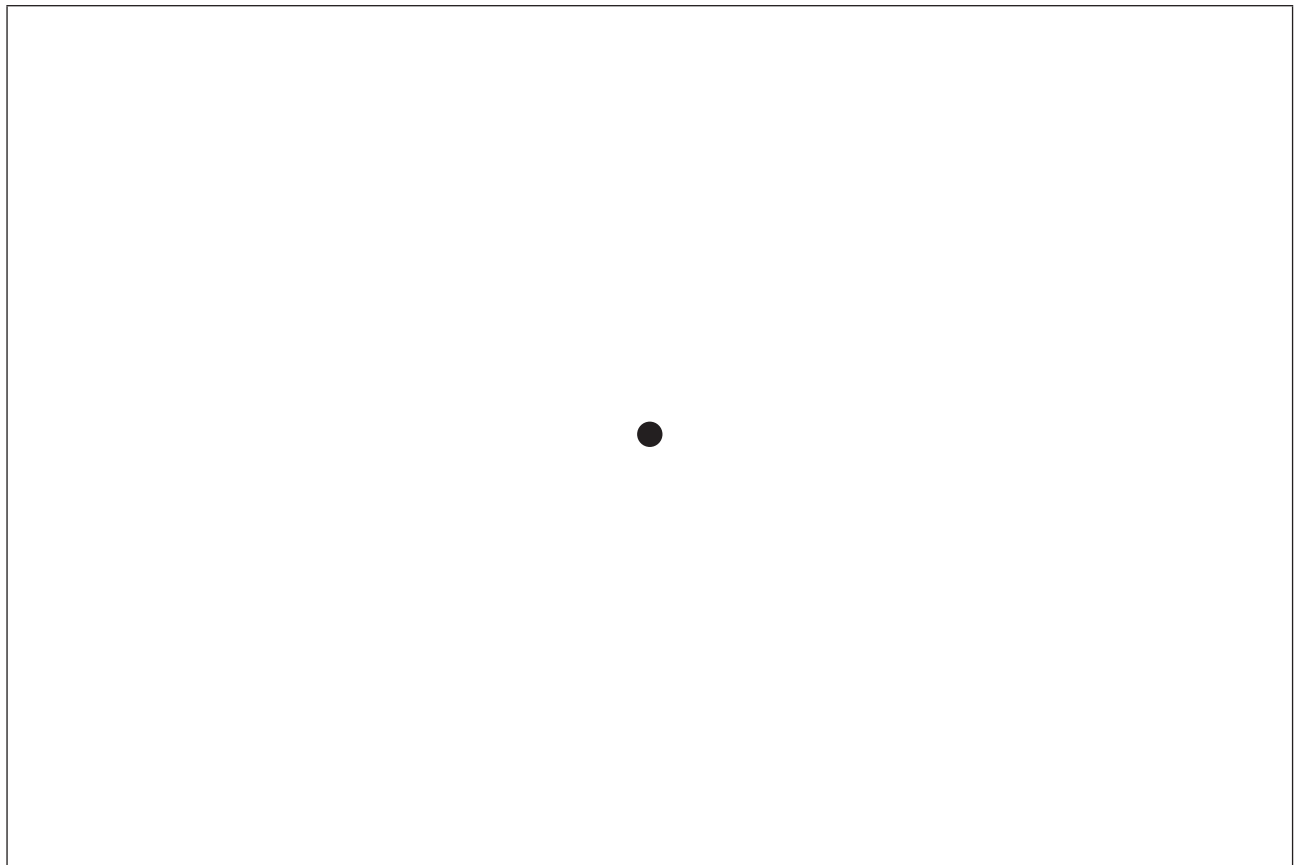


(Question A2 continued)

- (b) After the stone in (a) hits the water surface it rapidly reaches a terminal speed as it falls through the water. The stone leaves Fiona's hand at time $t=0$. It hits the water surface at t_1 and it comes to rest at the bottom of the water at t_2 . Using the axes below, sketch a graph to show how the speed v of the stone varies from time $t=0$ to just before $t=t_2$. (There is no need to add any values to the axes.) [3]



- (c) Draw and label a free-body diagram representing the forces acting on the stone as it falls through the water at its terminal speed. [2]



(Question B2 continued)

Part 2 Momentum

- (a) Define *linear momentum*. [1]

.....

.....

- (b) State the law of conservation of momentum. [2]

.....

.....

.....

.....

- (c) Far from any massive object, a space rocket is moving with constant velocity. The engines of the space rocket are turned on and it accelerates by burning fuel and ejecting gases. Discuss how the law of conservation of momentum relates to this situation. [3]

.....

.....

.....

.....

.....

.....

(This question continues on the following page)



(Question B2, part 2 continued)

- (d) Jane and Joe are two ice skaters initially at rest on a horizontal skating rink. They are facing each other and Jane is holding a ball. Jane throws the ball to Joe who catches it. The speed at which the ball leaves Jane, measured relative to the ground, is 8.0 m s^{-1} . The following data are available.

Mass of Jane = 52 kg

Mass of Joe = 74 kg

Mass of ball = 1.3 kg

Use the data to calculate the

- (i) speed v of Jane relative to the ground immediately after she throws the ball. [2]

.....

.....

.....

.....

- (ii) speed V of Joe relative to the ground immediately after he catches the ball. [2]

.....

.....

.....

.....

